Project Design Phase

FitBuddy – AI Fitness Plan Generator

1. Introduction

The project design phase is an essential step in the software development life cycle. In this phase, the system architecture, components, and interactions between different modules are defined. Proper design ensures that the system is scalable, efficient, and easy to maintain.

For the FitBuddy project, the design phase focuses on creating a structured architecture that integrates the frontend interface, backend API, artificial intelligence model, and database system. The objective is to ensure smooth communication between these components so that users can easily generate personalized workout plans.

The system follows a modular design approach where each component performs a specific task. This structure improves maintainability and allows future enhancements without affecting the entire system.

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2. System Architecture

The FitBuddy application follows a **three-tier architecture** consisting of the following layers:

1. Presentation Layer (Frontend)
2. Application Layer (Backend)
3. Data Layer (Database)

Architecture Overview

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User Interface (Frontend)
        ↓
FastAPI Backend
        ↓
AI Model + Database
```

This architecture ensures separation of responsibilities, making the system easier to manage and scale.

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3. Frontend Design

The frontend is responsible for interacting with the user and displaying the generated workout plans.

Technologies Used

- HTML
- CSS
- Jinja2 Templates

Key Interface Components

1. Home Page (index.html)

The homepage contains a form where users can input their fitness information.

The form collects the following details:

- Name
- Age
- Weight
- Fitness goal
- Workout intensity

Once the form is submitted, the data is sent to the backend for processing.

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2. Result Page (result.html)

The result page displays the generated workout plan and nutrition tips.

Features of this page include:

- Display of the AI-generated workout plan
- Nutrition advice
- Feedback form to modify workout plans
- Option to generate a new workout plan

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3. Admin Dashboard (all_users.html)

The admin dashboard allows administrators to view all registered users and their workout information.

The page displays:

- User name
- Age
- Weight

- Fitness goal
- Workout intensity

This information is retrieved from the database and displayed in a structured table format.

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4. Backend Design

The backend is responsible for handling user requests, communicating with the AI model, and managing database operations.

Backend Framework

The FitBuddy application uses **FastAPI** as the backend framework.

FastAPI is chosen because it offers:

- High performance
- Automatic API documentation
- Easy integration with Python-based AI libraries

API Endpoints

The backend includes several API routes to handle different functionalities.

1. Home Route

`/`

This route loads the homepage and displays the user input form.

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2. Generate Workout Plan

`/generate`

This route processes the user input and sends the information to the AI model to generate a workout plan.

Steps involved:

1. Receive user input from the form
2. Send prompt to AI model
3. Generate workout plan
4. Generate nutrition tip

5. Save data in database
6. Display results on the result page

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3. Submit Feedback

`/submit-feedback`

This route allows users to update their workout plan.

Steps involved:

1. Receive user feedback
2. Send feedback to AI model
3. Generate updated workout plan
4. Store updated plan in database
5. Display the new workout plan

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4. View All Users

`/view-all-users`

This route loads the admin dashboard showing all stored user records.

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5. Database Design

The database is responsible for storing user information and workout plans.

The FitBuddy project uses **MongoDB** or **SQLite** depending on deployment requirements.

User Collection/Table

Stores user details.

Fields include:

- user_id
- name
- age
- weight
- goal
- intensity

Workout Plan Collection/Table

Stores generated workout plans.

Fields include:

- workout_id
- user_id
- original_plan
- updated_plan

This structure allows the system to maintain both the original and updated workout plans.

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6. AI Model Integration

Artificial intelligence plays a crucial role in the FitBuddy system. The AI model is responsible for generating personalized workout plans and nutrition tips.

The backend sends structured prompts to the AI model that include the user's fitness details. Based on the prompt, the AI model generates a response containing a workout routine.

The generated content typically includes:

- Daily exercises
- Warm-up routines
- Workout sets and repetitions
- Cooldown suggestions
- Nutrition advice

To ensure reliability, fallback responses are implemented in case the AI API fails.

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7. Security Considerations

Several security measures are implemented in the system:

- API keys are stored in environment variables
- Sensitive data is not stored in the frontend
- Database access is restricted to backend services
- Input validation is applied to prevent invalid data submission

These measures help maintain system security and protect user data.

8. Advantages of the Design

The system design offers several advantages:

- Modular architecture for easy maintenance
- Clear separation between frontend, backend, and database
- Scalable structure for future improvements
- AI-powered automation of workout planning
- User-friendly interface for accessibility

9. Summary

The project design phase provides a detailed blueprint of the FitBuddy system. By defining the architecture, components, and interactions between modules, the design ensures that the application can be implemented efficiently and maintained effectively.

The structured architecture also allows future enhancements such as fitness tracking, user authentication, and mobile application integration.